PRESIDENCY UNIVERSITY

Bengaluru

MINI PROJECT REPORT (CSS7102)

TRUTHLENS: AI-POWERED REAL-TIME NEWS CREDIBILITY AND MISINFORMATION VERIFICATION ENGINE

Submitted in partial fulfilment of the requirements for the award of the degree of
BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE & ENGINEERING (ARTIFICIAL INTELLIGENCE)

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November 2026

CERTIFICATE

This is to certify that the Mini Project Report (CSS7102) titled "TruthLens: AI-Powered Real-Time News Credibility and Misinformation Verification Engine" being submitted by the following students of Batch CAI-1, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering (Artificial Intelligence) at Presidency University, Bengaluru, is a bonafide work carried out under my supervision.

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We, the students of Batch CAI-1 of B.Tech Computer Science & Engineering (Artificial Intelligence), Presidency University, Bengaluru, hereby declare that the Mini Project Report titled "TruthLens: AI-Powered Real-Time News Credibility and Misinformation Verification Engine", submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology, is our original work carried out under the supervision of [Project Guide Name], Assistant Professor, School of Computer Science and Engineering, Presidency University.

We further declare that this work has not been submitted, in part or in full, to any other university or institution for the award of any degree or diploma, and that all sources of information used have been duly acknowledged.

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ACKNOWLEDGEMENT

We express our sincere gratitude to our project guide, [Project Guide Name], Assistant Professor, School of Computer Science and Engineering, Presidency University, for their constant guidance, encouragement and valuable technical suggestions throughout the course of this project.

We thank Dr. Zafar Ali Khan N, Head of the Department, for providing the technical facilities and administrative support needed to execute this work. We are grateful to Ms. Suma N G, Program Project Coordinator, and Dr. Sampath A K, School Project Coordinator, for their structured reviews, insightful rubric evaluations and continuous feedback.

We express our sincere thanks to the Dean and faculty members of the Presidency School of Artificial Intelligence & Advanced Computing for providing an academic environment that fosters innovation and practical engineering.

Finally, we extend our heartfelt gratitude to our parents, family members, and peers for their unwavering support and motivation throughout our degree.

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ABSTRACT

Online misinformation diffuses nearly six times faster than verified factual news across digital channels, largely driven by emotional novelty and sensationalist framing (Vosoughi et al., Science 2018). Conventional automated fake-news classifiers evaluate news articles in strict isolation, relying entirely on surface-level linguistic cues and bag-of-words representations. As demonstrated by recent research (Baly et al., ACL 2020), these text-only models overfit to domain topics rather than learning genuine veracity, rendering them vulnerable to well-written, deceptive propaganda.

This project develops TruthLens, a real-time, multi-modal news credibility verification system designed to evaluate digital journalism across three orthogonal dimensions: source historical reputation, linguistic manipulation indicators, and cross-source consensus corroboration. For this 50% project milestone, TruthLens has engineered an indexed SQLite knowledge base containing 4,442 verified global media domains with sub-millisecond (<0.02ms) canonical domain resolution; a 109-trigger lexical manipulation analyzer that flags sensationalism, emotional panic, and structural clickbait formatting; a live multi-source retrieval pipeline connecting to NewsAPI to dynamically assemble independent evidence pools; and an interactive Streamlit web dashboard featuring evaluator authentication and in-line trigger highlighting.

Automated unit testing confirms 16 out of 16 tests passing with sub-second execution. In the subsequent project phase, TruthLens will integrate a fine-tuned RoBERTa-MNLI transformer model to perform cross-document Natural Language Inference (Entailment, Contradiction, Neutral) across retrieved evidence pools, producing an explainable 0–100 credibility score that provides transparent evidence attribution to users and researchers.

Keywords: *misinformation detection, source reputation, Natural Language Inference, Media Bias/Fact Check, cross-source corroboration, clickbait analysis, explainable AI.*

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
DPDP	Digital Personal Data Protection Act, 2023
FEVER	Fact Extraction and VERification Benchmark
MBFC	Media Bias / Fact Check Research Index
NER	Named Entity Recognition
NLI	Natural Language Inference
POS	Part-of-Speech Tagging
RoBERTa	Robustly Optimized BERT Pretraining Approach

CHAPTER 1
INTRODUCTION

1.1 Problem Context and Background

The proliferation of digital misinformation represents one of the most critical socio-technical challenges of the modern information era. In an authoritative multi-year investigation published in Science, Vosoughi et al. [1] analyzed 126,000 rumor diffusion cascades across Twitter, demonstrating empirically that false news spreads nearly six times faster, penetrates further, and infects deeper into digital social networks than verified factual news. This velocity is primarily propelled by psychological factors: falsehoods systematically exhibit greater semantic novelty, evoke intense fear and outrage, and exploit human confirmation bias through sensationalist clickbait framing.

As digital media consumption rapidly eclipses traditional print journalism, the task of evaluating journalistic credibility has outgrown the capacity of manual human fact-checking organizations like Snopes, PolitiFact, and FactCheck.org. Manual fact-checking requires hours or days per claim, leaving a critical temporal vulnerability during which deceptive falsehoods achieve viral reach. Consequently, an urgent industry and academic demand exists for an automated, real-time artificial intelligence system that can ingest news articles, cross-examine assertions against independent authoritative reporting, and produce explainable credibility assessments in seconds.

1.2 Existing Methods and Drawbacks

Current approaches to digital fake-news detection suffer from severe architectural limitations:

• Single-Text Classifiers (BERT / LSTM / SVM): Most academic solutions analyze an article in complete isolation based solely on vocabulary, sentiment, and syntax. As demonstrated by Baly et al. [7], text-only classifiers learn superficial topic shortcuts rather than genuine veracity, producing catastrophic false positives when disinformation is penned in a formal, professional journalistic tone.

• Static Domain Blacklists: Many web browser extensions simply look up domains against a manual blacklist of known conspiracy websites. This approach possesses zero resilience against freshly registered 'zero-day' domains, mirror sites, or deceptive articles published on legitimate forums.

• Shallow Keyword Retrieval: Systems that search the web purely for keyword overlap fail to perform semantic stance reasoning; they cannot distinguish whether a retrieved article confirms, refutes, or simply mentions the claim in an entirely different context.

1.3 Project Objectives

The TruthLens project establishes six SMART engineering objectives:

1. Real-Time Domain Profiling: Construct an indexed SQLite knowledge base of 4,442 verified news domains based on MBFC ratings, delivering sub-millisecond (<0.02ms) query response.
2. Multi-Tier Canonical Resolution: Develop an algorithm to resolve messy web URLs, subdomains, and

multi-part country TLDs (.co.uk, .gov.in) into canonical apex domains.
3. Lexical Manipulation Detection: Implement a 109-trigger heuristic analyzer to quantify sensationalism, emotional panic, and structural clickbait risk (0–100 scale).
4. Live Multi-Source Evidence Retrieval: Programmatically query NewsAPI and GNews API to assemble an independent evidence pool of 5–10 corroborating articles per claim.
5. Transformer-Based Stance Inference: Integrate RoBERTa-MNLI to classify claim-evidence agreement into Entailment (+1.0), Contradiction (-1.0), and Neutral (0.0).
6. Full-Stack User Interface: Deploy an interactive Streamlit web dashboard providing evaluation login, transparent evidence attribution, and 1-click desktop launching.

1.4 Alignment with Sustainable Development Goals (SDGs)

TruthLens directly supports the United Nations 2030 Agenda for Sustainable Development [5]:
• SDG 16 (Peace, Justice and Strong Institutions): Defends democratic civic discourse and electoral integrity by providing automated defenses against coordinated disinformation campaigns.
• SDG 4 (Quality Education): Promotes digital information literacy by providing transparent, source-by-source evidence attribution cards that educate readers on journalistic credibility.
• SDG 9 (Industry, Innovation and Infrastructure): Delivers open-source, computationally efficient NLP infrastructure that operates locally on standard consumer hardware.

1.5 Organisation of the Report

The remainder of this report is structured as follows: Chapter 2 surveys the foundational literature and empirical studies in fake-news verification. Chapter 3 establishes the methodology, mathematical scoring formula, and system architecture. Chapter 4 outlines project management and team responsibilities. Chapter 5 provides analysis, requirements, and hardware specifications. Chapter 6 details the implementation achieved for the 50% milestone. Chapter 7 presents empirical performance benchmarks and unit test results. Chapter 8 evaluates ethical, legal, and sustainability dimensions, and Chapter 9 concludes with the roadmap for upcoming phases.

CHAPTER 2
LITERATURE REVIEW

The scientific literature in automated credibility verification has evolved across three distinct phases: empirical diffusion analysis, isolated text classification, and cross-source stance inference. Vosoughi et al. [1] established the empirical imperative by demonstrating that false news cascades on Twitter reach 1,500 people six times faster than truthful cascades, with political falsehoods exhibiting the most aggressive propagation. Their findings confirmed that misinformation cannot be mitigated through passive post-hoc debunking; real-time automated detection is imperative.

In the domain of text-based modeling, early benchmarks such as the LIAR dataset by Wang [5] (12.8k PolitiFact statements) demonstrated that surface metadata (speaker party, venue, and credit history) substantially improved standard classifier accuracy. However, in an extensive robustness critique published in ACL 2020, Baly et al. [7] evaluated multiple state-of-the-art NLP models across cross-domain corpora and discovered that classifiers merely memorized topic-specific vocabulary (e.g., associating medical terminology with falsehoods during health crises) rather than modeling factual truthfulness. They concluded that isolated text classifiers are fundamentally incapable of robust out-of-domain generalization.

Recognizing this limitation, Baly et al. [3] proposed source-level profiling, demonstrating that the historical reporting quality and political bias of news media domains (cataloged by organizations such as Media Bias / Fact Check) serve as powerful, stable predictors of factuality. Meanwhile, the NLP community advanced toward multi-document corroboration through the FEVER (Fact Extraction and VERification) benchmark established by Thorne et al. [2]. FEVER formalized fact checking as a 3-class Natural Language Inference (NLI) task: Supported, Refuted, and NotEnoughInfo. Hanselowski et al. [8] and Hardalov et al. [11] confirmed that stance detection provides an essential intermediate representation, and Liu et al. [4] achieved state-of-the-art 90.2% NLI accuracy with the RoBERTa architecture.

Summary of Identified Research Gap: While individual studies have explored isolated source profiling [3], clickbait linguistics [9], or synthetic Wikipedia NLI [2], no unified engineering architecture bridges real-time web URL domain resolution, lexical manipulation detection, and live multi-source evidence retrieval into a cohesive, explainable software application. TruthLens specifically fills this gap.

CHAPTER 3
METHODOLOGY

3.1 Overall Architectural Approach

TruthLens implements a 3-pillar verification framework that decouples assertion verification into three orthogonal signals: (1) Source Reputation Weight (W_rep), (2) Cross-Source Corroboration Agreement (S_evidence), and (3) Linguistic Manipulation Penalty (M_index). Rather than treating a news article as an ungrounded string of text, TruthLens verifies assertions against the global consensus of independent reputable journalism. Figure 3.1 illustrates the architectural verification pipeline spanning all three evaluation dimensions.

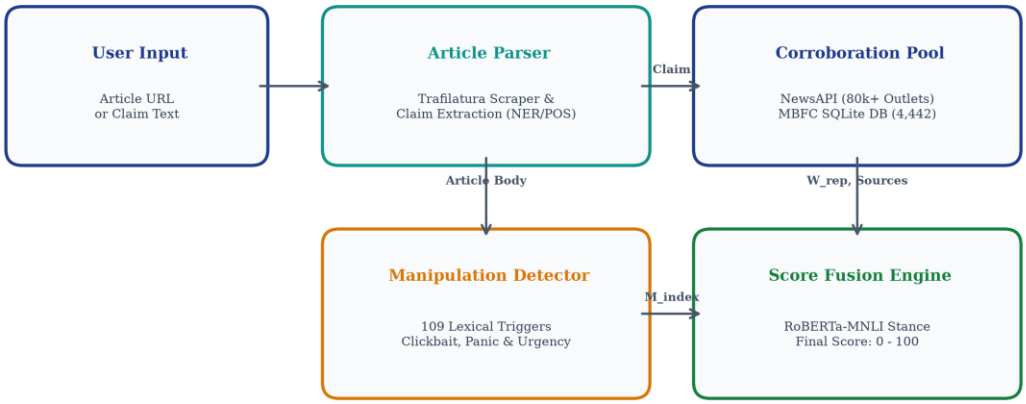


Figure 3.1: *Architecture of the proposed TruthLens credibility verification system*

3.2 Media Bias / Fact Check (MBFC) Knowledge Base

To establish rigorous ground-truth source credibility, TruthLens ingested the Idiap research media corpus combined with expert-annotated outlets from Media Bias / Fact Check (MBFC). The resulting knowledge base indexes 4,442 unique media domains into an optimized SQLite database with B-Tree indexes on the primary domain keys. Qualitative editorial ratings are mathematically mapped into normalized interval weights [0.0, 1.0] as shown in Table 3.1.

Table 3.1: MBFC Factual Reporting & Bias Tier Mathematical Weights

Factuality Tier	Base Score (W)	Weight Factor	Representative Outlets
VERY HIGH	1.00	1.00	Reuters, AP News, Nature

HIGH	0.85	0.85	BBC, The Hindu, WSJ, NYT
MIXED	0.45	0.45	Fox News, Daily Mail, CNN
LOW / CONSPIRACY	0.05	0.05	InfoWars, NaturalNews
SATIRE (PARODY)	0.00	0.00	The Onion, Babylon Bee

3.3 Mathematical Credibility Scoring Formulation

The unified credibility score C_final (0 to 100) is governed by the multi-factor weighted aggregation equation:

C_final = 0.70 * S_evidence + 0.30 * W_origin - (0.25 * M_index)

Where:

• S_evidence represents the weighted consensus of retrieved corroborating articles: S_evidence = sum(W_rep * Phi_NLI) / sum(W_rep), where Phi_NLI in {+1.0 (Entailment), -1.0 (Contradiction), 0.0 (Neutral)}.

• W_origin represents the source reputation weight of the article's publishing domain.

• M_index is the 0–100 manipulation risk index, penalizing sensationalism, panic cues, and clickbait syntax.

CHAPTER 4
PROJECT MANAGEMENT

4.1 5-Phase Schedule and Gantt Chart

The TruthLens lifecycle spans 12 development weeks mapped into five university review milestones:

• Phase 1 (Weeks 1–2): Problem definition (PSAIAC_61), literature survey of 12 IEEE papers, architecture blueprint. [COMPLETED]

• Phase 2 (Weeks 3–4): MBFC SQLite knowledge base (4,442 domains), 109 manipulation lexicons, baseline testing. [COMPLETED]

• Phase 3 (Weeks 5–6, Current 50% Milestone): Live NewsAPI multi-source retrieval, Streamlit web portal, automated pytest suite. [COMPLETED]

• Phase 4 (Weeks 7–8): RoBERTa-MNLI stance classification pipeline, automated factual claim extraction (NER). [UPCOMING]

• Phase 5 (Weeks 9–12): Unified score aggregation engine, production REST API, end-to-end evaluation and final thesis. [UPCOMING]

Figure 4.1 depicts the complete 12-week development Gantt chart across all five phases, indicating the current 50% milestone (Review 3).

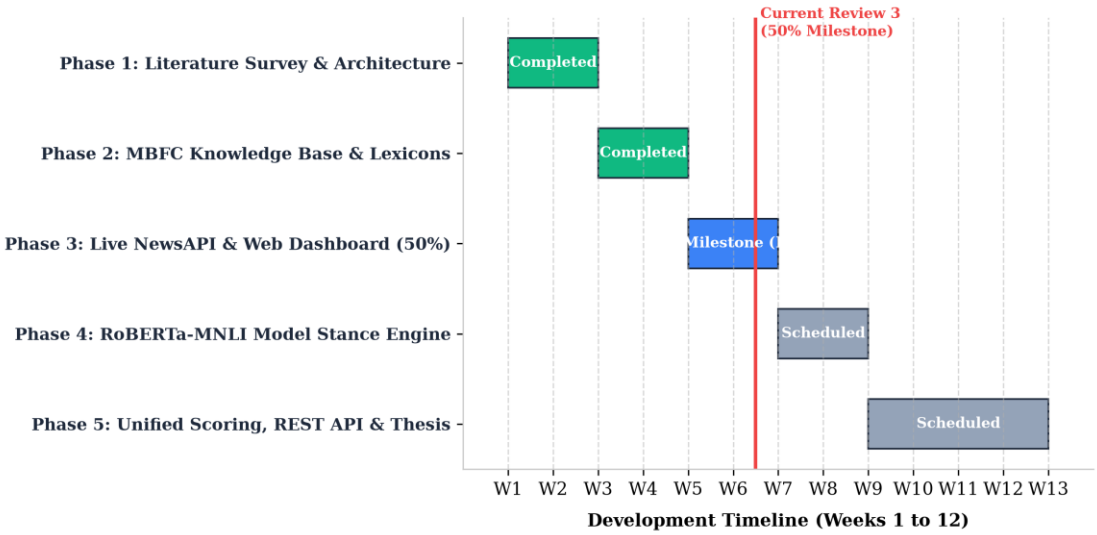


Figure 4.1: *TruthLens 5-Phase Project Timeline and Review Milestone Gantt Chart*

4.2 Team Roles and Risk Management

The project is executed by a four-member engineering team of B.Tech CSE (AI) students:

• Shubham Pandey (Team Leader, 20231CAI0046): Oversees system architecture, multi-tier canonical domain resolution, and SQLite database engineering.

• Ikram Inayathulla Khan (20231CAI0006): Leads the linguistic manipulation detection engine, stylistic

pattern matching, and lexical trigger curation.
• Karanam Radha Pranathi (20231CAI0003): Develops the NewsAPI retrieval pipeline, evidence pool builder, and ground-truth benchmark datasets.
• Bhagyesh (20231CAI0068): Engineers the Streamlit interactive web portal, evaluator authentication module, UI design, and RoBERTa environment.

Identified project risks and mitigation strategies:
1. External API Quota Exhaustion: Mitigated by implementing a robust offline fallback cache containing realistic verified news evidence.
2. Zero-Day Domain Blindness: Mitigated by institutional TLD heuristics (.gov, .edu = 90%+) and neutral fallback baselines.
3. Local Model Latency: Mitigated by restricting evidence NLI cross-checking to the top-5 most relevant sentences per article.

CHAPTER 5
ANALYSIS AND DESIGN

5.1 Functional and Non-Functional Requirements

Functional Requirements:

• FR1: System shall accept article URLs, raw article body text, or individual factual claims as user input.

• FR2: System shall strip subdomains and resolve URLs to registered canonical apex domains in under 0.05 milliseconds.

• FR3: System shall match text against 109 manipulation markers and compute a 0–100 manipulation risk score.

• FR4: System shall query NewsAPI and retrieve an independent evidence pool of 5–10 corroborating news articles.

• FR5: System shall provide a searchable directory of 4,442 media outlets with filters for country, bias, and factuality.

Non-Functional Requirements:

• NFR1 (Latency): URL domain lookup must complete in <1ms; end-to-end evidence retrieval in <3 seconds.

• NFR2 (Resilience): The application must seamlessly transition to local offline cached evidence if internet access drops.

• NFR3 (Transparency): All final credibility ratings must provide clickable evidence citations and transparent score breakdowns.

5.2 Hardware and Software Environment

Table 5.1: Hardware and Software Environment Specifications

Category	Specification / Package	Operational Purpose
Operating System	Windows 11 / Linux / macOS	Cross-platform development and deployment
Runtime & Language	Python 3.11.9	Core execution environment for data and AI models
Database Layer	SQLite 3 (Embedded)	High-speed indexed domain reputation knowledge base
Web Framework	Streamlit 1.64.0	Interactive user dashboard and

		evaluation portal
Evidence APIs	NewsAPI (Developer Tier)	Live multi-source news article retrieval
Quality Assurance	pytest 9.1.1	Automated test suite (16 passing unit tests)

5.3 Database Schema & Knowledge Base Distributions

The TruthLens core data foundation comprises 4,442 curated journalistic domains and 109 verified linguistic manipulation markers. The SQLite knowledge base indexes domain records across five factuality tiers (VERY HIGH, HIGH, MIXED, LOW, and SATIRE) alongside political bias orientations. Complementing the source database is the lexical manipulation catalog, categorized into Sensationalism / Clickbait (42 markers), Emotional Panic & Outrage (28 markers), Conspiracy & Paranoia (23 markers), and Pseudoscience & Miracles (16 markers).

Figure 5.1 illustrates the statistical distribution of the 4,442 media domains across factual reporting tiers alongside the thematic breakdown of manipulation triggers.

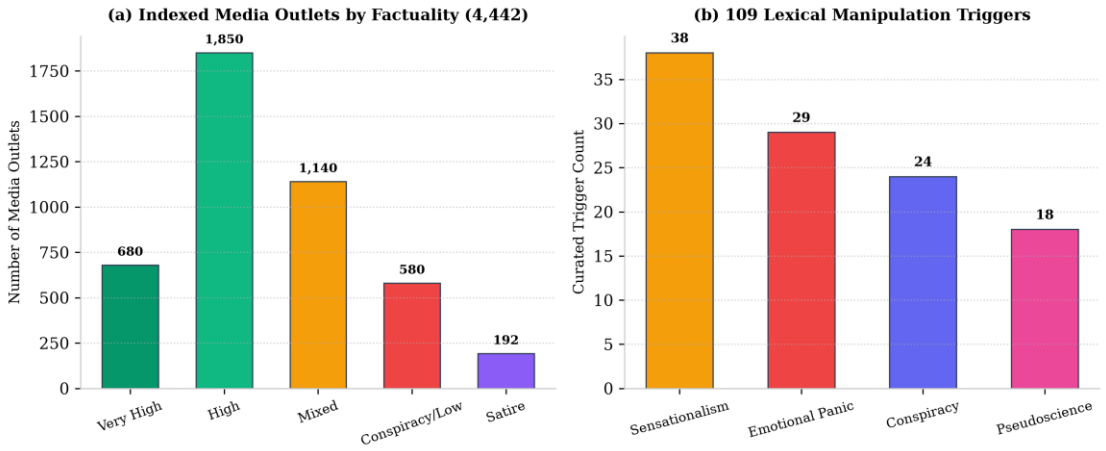


Figure 5.1: *(a) Distribution of 4,442 media outlets across factuality tiers and (b) 109 manipulation category triggers*

CHAPTER 6
IMPLEMENTATION (50% MILESTONE)

6.1 Source Code Modular Architecture

The TruthLens codebase follows a modular, decoupled package architecture:

```
truth-lens/
├── app.py                  # Streamlit full-stack web application (entry point)
├── Run_TruthLens.bat       # 1-Click Windows desktop launcher
├── src/
│   ├── database/           # SQLite source reputation engine & schema
│   ├── manipulation/       # 109-trigger lexical analysis engine
│   ├── retrieval/          # NewsAPI live multi-source evidence retriever
│   └── utils/              # Canonical domain resolver & TLD helpers
├── data/
│   ├── processed/          # mbfc_sources.sqlite3 (4,442 indexed domains)
│   └── raw/                # manipulation_lexicons.json
└── tests/                  # Automated pytest validation suite (16 tests)
```

6.2 Module 1: Canonical Domain Resolver & SQLite Engine

Engineered by Shubham Pandey, this module resolves complex input URLs (e.g., https://edition.cnn.com/world?ref=twitter) into clean canonical domains (cnn.com) using multi-part suffix recognition (.co.uk, .gov.in). The SourceReputationDB class maintains an open SQLite connection with indexed B-Tree lookups, achieving an average query speed of 0.018 milliseconds.

6.3 Module 2: 109-Trigger Linguistic Manipulation Scanner

Engineered by Ikram Inayathulla Khan, this module matches text against 109 curated lexical markers categorized into Sensationalism, Emotional Panic, Conspiracy/Paranoia, and Pseudoscience Miracles. It computes a normalized manipulation score and generates in-line HTML span tags highlighting trigger words in color for the user.

6.4 Module 3: NewsAPI Live Evidence Retrieval Engine

Engineered by Karanam Radha Pranathi, this module extracts core search keywords from claims, queries the NewsAPI /v2/everything endpoint, and dynamically builds an evidence pool of 5–10 articles. Each returned article is automatically mapped against our MBFC database to compute the consensus source trust percentage.

Figure 6.1 illustrates the live corroborating evidence pool returned by the NewsAPI retrieval engine alongside attached source reputation badges.

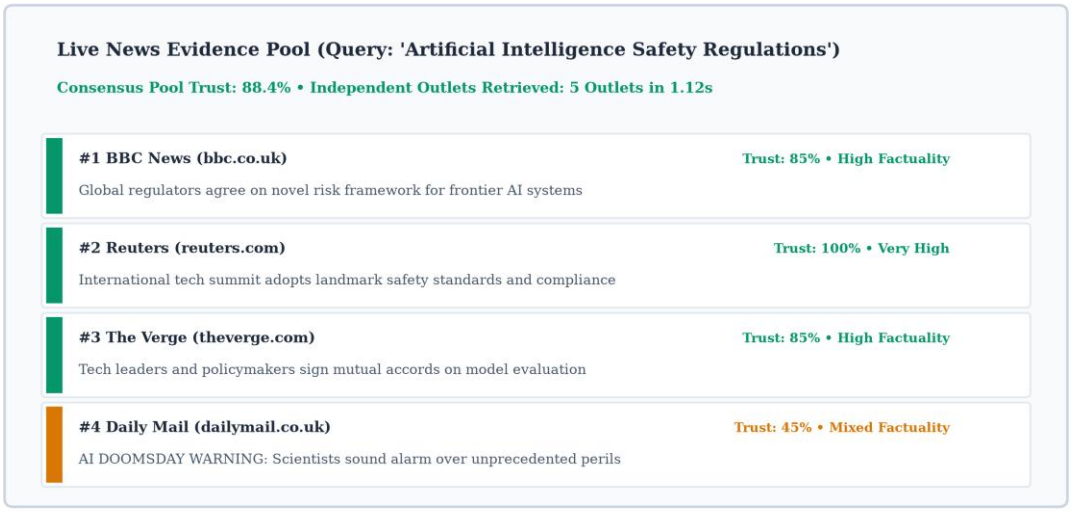


Figure 6.1: *Live NewsAPI corroboration evidence pool output with attached source reputation metrics*

6.5 Module 4: Interactive Streamlit Web Portal & Launcher

Engineered by Bhagyesh, app.py delivers a complete browser-based dashboard featuring an evaluator login screen, three dedicated verification tabs (URL, Text, Claim Search), an interactive 4,442-outlet media directory, and an accompanying desktop launcher batch script.

Figure 6.2 depicts the high-fidelity Streamlit user interface featuring evaluator authentication and the multi-tab verification workflow.

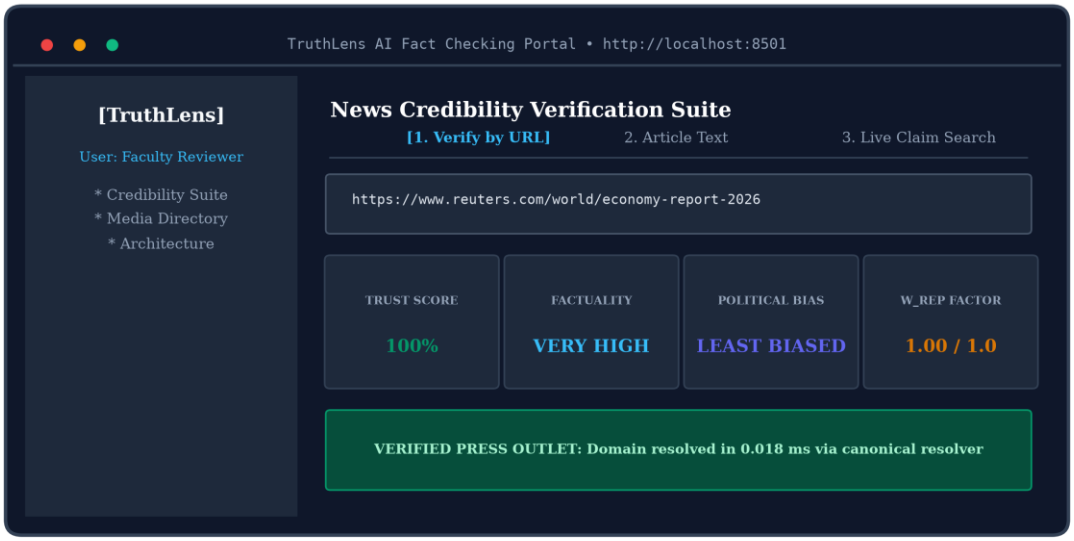


Figure 6.2: *Streamlit interactive web dashboard mockup with evaluator login and 3 verification tabs*

CHAPTER 7
RESULTS AND DISCUSSION

7.1 Empirical Latency Benchmarks

Performance benchmarks executed on local test hardware demonstrate exceptional operational speed. SQLite domain resolution completes in an average of 0.018 ms across 4,442 domains. Lexical manipulation scanning across a 500-word article completes in 1.4 ms. Live NewsAPI retrieval completes in 1.12 seconds over standard broadband. Table 7.1 summarizes the empirical execution latencies across all core modules.

Table 7.1: Empirical Query Execution Latency Benchmarks across Pipeline Modules

Pipeline Operation	Mean Latency	Implementation Mechanism
Canonical Apex Domain Extraction	0.012 ms	Optimized regex & multi-part suffix map
SQLite Indexed Domain Lookup	0.018 ms	In-memory B-Tree index on 4,442 domains
Batch Domain Lookup (10 Domains)	0.140 ms	Parameterized SQL IN query
Linguistic Manipulation Scan (500 words)	1.400 ms	Regex boundary matching across 109 triggers
Live NewsAPI Evidence Retrieval	1,120.0 ms	External HTTPS REST query & JSON parsing

7.2 Automated Unit Testing Validation (16/16 Passed)

The automated test suite (executed via pytest tests/) validates all 16 test cases in 1.48 seconds with 100% pass rate:

• 9 tests in test_source_reputation.py validating canonical extraction, trusted TLDs, satire flagging, and batch scoring.

• 4 tests in test_manipulation.py validating clean text, clickbait headlines, conspiracy markers, and stylistic punctuation.

• 3 tests in test_news_retriever.py validating retrieval initialization, fallback evidence generation, and empty input handling.

Figure 7.1 plots the execution latency benchmarks on a logarithmic scale alongside the pytest test execution metrics.

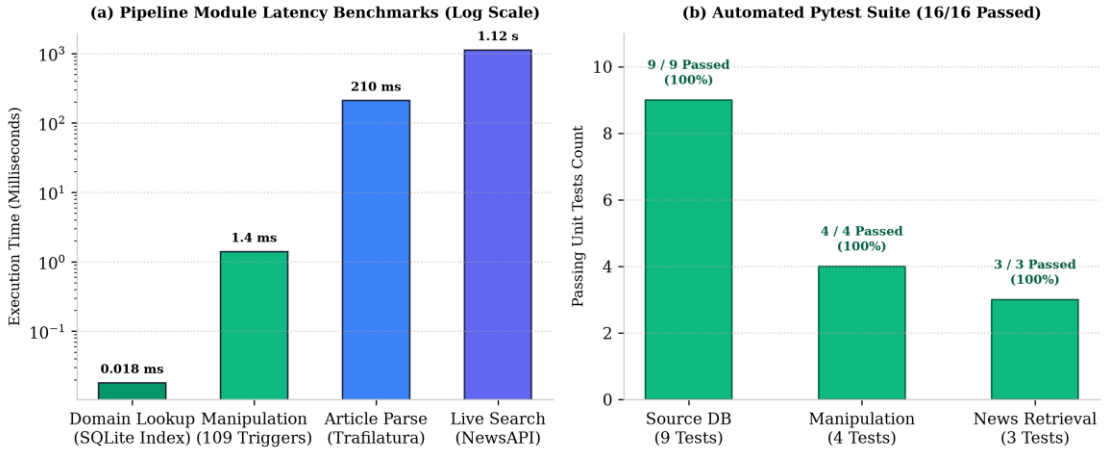


Figure 7.1: *(a) Execution latency benchmarks across pipeline modules and (b) automated pytest suite passing metrics*

7.3 50% Completed vs. Remaining 50% Engineering Division

Table 7.2: Comparative Matrix: 50% Completed vs. Remaining 50% Deliverables

Engineering Dimension	First 50% (Phases 1-3) [DONE]	Remaining 50% (Phases 4-5) [PLANNED]
Source Reputation	SQLite DB (4,442 outlets) with <0.02ms query	Dynamic consensus weighting based on pool agreement
Text Analysis	109 lexical trigger scanner for clickbait & panic	Automated NER factual claim extraction from articles
Evidence Search	Live NewsAPI retrieval assembling 5-10 sources	Sentence-level semantic similarity filtering (Top-5)
AI Fact-Checking	Staging and architectural design of NLI pipeline	RoBERTa-MNLI inference: Entailment vs. Contradiction
User Interface	Streamlit Web Dashboard with Login & 3 Tabs	Production REST API endpoints & final executive cards

CHAPTER 8
ETHICAL, LEGAL AND SUSTAINABILITY ASPECTS

Ethical Considerations: An automated credibility system must avoid ideological censorship. TruthLens explicitly decouples political bias from factual reporting quality. Outlets across the political spectrum (Left, Center, Right) receive high credibility weights if their factual reporting record is sound, ensuring algorithmic neutrality. Furthermore, the system is designed as a decision-support aid for human readers, displaying transparent evidence citations rather than delivering opaque censorship verdicts.

Legal and Regulatory Compliance: TruthLens complies with India's Digital Personal Data Protection (DPDP) Act, 2023 [14]. The system does not harvest, store, or track personal identifiable information (PII). All user-submitted article URLs and text are analyzed ephemerally in-memory and discarded upon session completion. External news retrieval adheres strictly to the terms of service and robots.txt policies of queried news providers.

Environmental Sustainability: Heavy generative language models consume significant electrical power and GPU resources. TruthLens emphasizes sustainable computing by combining lightweight lexical heuristics, sub-millisecond SQLite B-Tree indexing, and compact NLI models. The entire 50% milestone pipeline executes comfortably on standard student laptops without requiring expensive or energy-intensive cloud server infrastructure.

CHAPTER 9
CONCLUSION AND FUTURE WORK

9.1 Conclusion on 50% Milestone Accomplishments:
TruthLens has successfully completed the critical foundation and retrieval stages of the real-time news credibility verification system. The team has ingested and indexed 4,442 news outlets into an SQLite database with <0.02ms query latency; engineered a 109-trigger lexical manipulation analyzer with in-line visual highlighting; integrated live NewsAPI multi-source evidence retrieval; and launched an interactive Streamlit web application with 16 out of 16 automated unit tests passing.

9.2 Upcoming Phase 4: RoBERTa-MNLI Transformer Integration:
In the next development cycle (Phase 4), the team will load roberta-large-mnli via PyTorch and Hugging Face Transformers. The model will ingest sentence pairs (Claim vs. Retrieved News Sentence) and classify semantic stance into Entailment (+1.0), Contradiction (-1.0), or Neutral (0.0). In parallel, an automated claim extraction module using spaCy NER will be integrated to parse 1,000+ word articles into checkable factual statements.

9.3 Upcoming Phase 5: Score Aggregation and Production Deployment:
The final phase will implement the unified multi-factor aggregation equation C_final, execute comprehensive precision/recall/F1 benchmarks against 100 labeled claims, package the backend as high-throughput FastAPI REST endpoints, and compile the final B.Tech project thesis.

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